① Aspect 12 (Most Central)

Use a multi-step procedure leading to a mathematical solution

drip-rate-qs2 typically requires students to:

- Use the given infusion drip-rate formula
- Carry out necessary unit handling (e.g. converting hours to minutes)
- Substitute numerical values
- Complete the calculation to obtain the final drip rate

The solution is therefore not achieved in a single step, but through a **clear, ordered, multi-step procedure**, making Aspect 12 the most central.

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② Aspect 5

Understand and utilise constructs based on definitions, rules, and formal systems

Students must understand and apply:

- The definitions of the variables in the formula (drop factor, volume, time, drip rate)
- Fractional structures and proportional relationships
- Fundamental rules linking time, rate, and quantity

This reflects students' ability to work within **formal mathematical structures and rules**.

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③ Aspect 7

Manipulate numbers and algebraic expressions

During problem solving, students need to:

- Substitute numerical values correctly
- Perform multiplication and division
- Manage denominators, units, and quantitative relationships

This aspect captures the **numerical and algebraic manipulation skills** that underpin the multi-step solution.

④ Aspect 3

Implement a given strategy to determine a mathematical solution

The strategy is effectively predetermined:

> Apply the specified formula step by step to calculate the drip rate.

Students are assessed on their ability to:

- Understand this given strategy
- Execute it accurately and consistently

The emphasis is on **implementation rather than strategy selection**.

⑤ Aspect 13

Use an understanding of the context to guide the mathematical solving process

Within the medical context, students must consider:

- The relationship between infusion time and drip rate
- Whether the final rate should be expressed as a whole number
- Whether the result lies within a realistic range for nursing practice

Contextual understanding helps **validate and refine the solution**, though it is not the main cognitive challenge.

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⑥ Aspect 8

Articulate a solution, showing and/or summarising intermediate results

In a complete response, students are usually expected to:

- Show key intermediate steps
- Clearly state the final drip rate, including appropriate units

Clear presentation of the solution process and result is part of the assessment, though not the central difficulty.

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⑦ Aspect 1

Perform a simple calculation

Each individual operation typically involves:

- Basic arithmetic

While accuracy is essential, the challenge lies in **organising and executing the sequence of steps**, not in the complexity of individual calculations.

① Aspect 12（最核心）

Use a multi-step procedure leading to a mathematical solution

📌 *drip-rate-qs2* 通常要求学生：

- 使用给定的输液滴速公式
- 先进行必要的单位处理（如将小时转化为分钟）
- 再代入数值完成计算
- 得到最终的滴速结果

👉 解题并非一步完成，而是一个**清晰、有顺序的多步骤计算过程**，因此 Aspect 12 是最核心的体现。

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② Aspect 5

Understand and utilise constructs based on definitions, rules and formal systems

📌 学生需要理解并运用：

- 公式中各变量的定义（滴因子、体积、时间、滴速）
- 分式结构与比例关系
- 时间与速率之间的基本数学规则

👉 题目考查的是对**正式数学结构与规则的理解和运用**，而非直观猜测。

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③ Aspect 7

Manipulate numbers and algebraic expressions

📌 在解题过程中，学生需要：

- 正确代入数值
- 进行乘除运算
- 处理分母、单位和数量关系

👉 这是典型的**数值与代数表达式操作能力**，为完成多步骤求解提供基础。

④ Aspect 3

Implement a given strategy to determine a mathematical solution

📌 题目已经隐含或明确给出解题策略：

> 使用指定公式 → 按步骤计算滴速

学生的任务在于：

- 理解这一既定策略
- 准确、稳定地执行它

👉 重点不在于“选择策略”，而在于**落实和执行给定策略。**

⑤ Aspect 13

Use an understanding of the context to guide the mathematical solving process

📌 在医疗情境中，学生需要考虑：

- 时间与滴速之间的现实关系
- 滴速是否需要取整数
- 结果是否符合护理实践的合理范围

👉 情境理解帮助学生**校正计算结果的合理性**，但并非解题的主要认知负担。

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⑥ Aspect 8

Articulate a solution, showing and/or summarising intermediate results

📌 在完整作答中，通常需要：

- 展示关键中间步骤
- 清楚给出最终滴速值（含单位）

👉 清晰表达计算过程与结果是评价的一部分，但不是本题的核心难点。

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⑦ Aspect 1

Perform a simple calculation

📌 单个运算步骤本身通常是：

- 基本四则运算

👉 虽然计算准确性重要，但**题目的挑战不在单次计算的复杂性**，因此该方面相关性较低。